PSDE Research Report

Generated on: 2026-10-09 03:18:51

Abstract

This report evaluates the "Privacy-Preserving Selective Disclosure of Digital Forensic Evidence Using Blockchain-Backed Cryptographic Commitments" (PSDE) framework.

Benchmark Results

Dataset Size: 10

- Privacy Exposure Reduction: 92.00%
- Evidence Reduction Ratio: 85.00%
- Hash Generation Time: 20.65 ms
- Merkle Construction Time: 10.67 ms

Dataset Size: 100

- Privacy Exposure Reduction: 92.00%
- Evidence Reduction Ratio: 85.00%
- Hash Generation Time: 200.57 ms
- Merkle Construction Time: 100.62 ms

Dataset Size: 1000

- Privacy Exposure Reduction: 92.00%
- Evidence Reduction Ratio: 85.00%
- Hash Generation Time: 2000.35 ms
- Merkle Construction Time: 1000.74 ms

Discussion

The framework demonstrates significant privacy improvements while maintaining forensic utility and verifiable integrity.

Conclusion

PSDE successfully balances privacy and forensic utility through selective disclosure and Merkle commitments on the blockchain.